



Nighttime Chlorophyll Fluorescence Imaging of Dark-Adapted Plants Using a Robotic Field Phenotyping Platform

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Abstract

Photosynthetic efficiency is increasingly recognized as an integration of plant responses to dynamic environments, establishing the need for data sets from both field trials and controlled environments. A robotic field scanner phenotyping platform at the University of Arizona is equipped with a high-throughput chlorophyll fluorescence imaging system capable of collecting data on field trials for genetic studies of a photosynthetic trait (F_v/F_m). A description of the fluorescence imaging system is provided in addition to methods for measurements across experimental field plots and a test to determine the impact of variable plant heights. The overall focus is on aspects of field applications of a chlorophyll fluorescence imaging system that differ from analogous systems in controlled environments.

Key words Field scanner, PSII, Chlorophyll fluorescence, Quantum efficiency, High-throughput phenotyping, Field phenotyping

1 Introduction

Chlorophyll fluorescence imaging has been widely applied as a non-contact plant phenotyping method for characterizing the activity of photosynthetic systems in indoor controlled environments [1]. Photosynthetic activity is a fundamental determinant of plant health and productivity and is the result of complex genetics. This provides opportunities for phenotype to genotype studies to guide breeding for improved crop varieties with enhanced photosynthetic traits [2]. There is increasing awareness of the importance of studying photosynthetic parameters in field environments and associated interest in chlorophyll fluorescence imaging systems that can be used as a tool in field phenotyping programs. While there are multiple benefits to controlled environment studies that can isolate particular variables of interest, the fact that photosynthesis efficiency is an integration of plant responses to dynamic

environments establishes the need for data sets from both field trials and controlled environments [3].

To date, studies using in-field automated phenotyping systems to assess photosynthesis in fluctuating light conditions have been rare [4]. Over half a decade ago, Murchie and Lawson (2013) predicted the incorporation of remote fluorescence measurements into automated high-throughput crop phenotyping methods and techniques. Here we describe a novel field imaging system and methods for chlorophyll fluorescence measurements of dark-adapted plants using the robotic field scanner system at the University of Arizona Maricopa Agricultural Center. The field fluorescence imaging system has been utilized in 2 different field trials to measure the maximum quantum efficiency of photosystem II (estimated as F_v/F_m) of dark-adapted plants for a genotyped bi-parental sorghum mapping population of 160 RILs planted in 2 replicates (320 plots) and a genotyped durum wheat diversity panel of 223 lines in 2 replicates and 2 irrigation treatments plus repeating checks (924 plots). These phenotyping scans are part of the first applications of a robotic semi-automated field phenotyping platform to obtain high-throughput chlorophyll fluorescence image data for genetic studies of a photosynthetic trait (F_v/F_m).

Quantification of chlorophyll *a* fluorescence is a useful indicator of the status of photosynthetic processes since emitted photons inform the relative rates of photochemical and non-photochemical quenching. Several reviews describe the process by which light energy absorbed by plant photosystems is dissipated as either photochemistry, heat, or re-emitted photons (fluorescence) [5–7]. The maximum quantum yield of the photosystem II (PSII) system can be determined by measuring fluorescence at known states: fully dark-adapted when PSII reaction centers are electrically neutral or open (F_0 minimal fluorescence) and following a saturating illumination when reaction centers cannot use the excitation energy as a result of charge separation and are closed (F_m maximum fluorescence) [7]. Together the fluorescence measurements at these two states can be used to determine the quantum efficiency of PSII photochemistry using the equation $F_v/F_m = (F_m - F_0)/F_m$ [6]. F_v/F_m values for plant leaves in the absence of any stress are close to 0.83 across species and indicate plant stress and overall metabolism that directly or indirectly reflects photosynthetic functions [4, 6, 8].

Incorporating chlorophyll fluorescence image data in high-throughput phenotyping programs in field environments has tremendous value and also challenges. In field studies, imaging platforms allow chlorophyll fluorescence measurements at the scale of whole plants and plant canopies and capture the variability among plants as well as within plants under natural conditions [9]. A complete system includes an illuminating light with sufficient intensity to saturate the photosystems of plant leaves over the

imaged area, a synchronized and robust camera fit with a bandpass filter, cooling capacity to accommodate the light source, and importantly a mechanism for moving the system across a field trial. The design and installation require engineering expertise and knowledge of the biophysics of fluorescence and photosynthesis parameters [9]. While commercial options for automated field imaging systems are currently limited, for example, PhenoVation [10] and Qubit Systems, it is likely that available options will increase in the near future.

The field scanner at the University of Arizona Maricopa Agricultural Center is a large robotic gantry system constructed and employed by the TERRA-REF project funded by the Department of Energy ARPA-E to reduce the phenotyping bottleneck that has slowed the improvement of crop varieties (www.terraref.org) (Fig. 1). The field scanner can cover a total of 0.75 hectares of field plots to collect high-resolution, high-frequency, and accurate data on phenotypic traits of interest for diverse sets of germplasm. The scanner system is semi-automated and can scan during day and night hours. It is capable of motion along three axes at velocities of 1.0 m/s on the long axis and 0.33 m/s on the short axis. An array of imaging and sensor systems are integrated with scan motion and data transfer systems, including a thermal infrared camera, hyperspectral imagers, Fraunhofer 3D laser systems, stereo RGB cameras, and NDVI and PRI analog sensors. Additionally, the field phenotyping platform includes a chlorophyll fluorescence imaging system that is a LemnaTec prototype instrument (www.lemnatec.com). A description of the fluorescence imaging system follows in addition to the basic methods for measurements across experimental field plots and a test to determine the impact of variable plant heights. The overall focus is on aspects of field applications of a chlorophyll fluorescence imaging system that differ from analogous systems in controlled environments.

2 Materials

The prototype imaging system is a novel deployment of a chlorophyll fluorescence instrument on a robotic platform in a field setting. The integrated imaging system includes a large panel of red light-emitting diodes (LED). It is operated at night to measure photosystem II activity on dark-adapted plants (Fig. 2). At an operating distance of 80 cm from the plant canopy, the area imaged per measurement is 1.1×0.7 m. The illuminating flash has a dominant wavelength in the range of 620–630 nm and sufficient intensity (up to $7000 \mu\text{mol}/\text{m}^2/\text{s}$ at 70 cm from plant target) to saturate the photosystem of plants. The system is modular and scalable with a controller that interfaces by Ethernet to the camera driver on one end and interfaces with other drivers to synchronize



Fig. 1 The field scanner at the University of Arizona Maricopa Agricultural Center

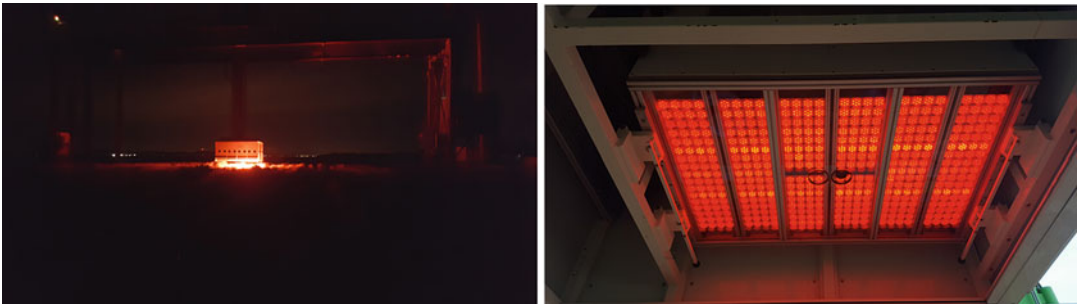


Fig. 2 The chlorophyll fluorescence imaging system includes an LED panel that provides a saturating flash of light over the imaged area

triggering of the flash. The camera is manufactured by Allied Vision (Manta G-235b) and is fit with a bandpass filter mounted on the camera objective in replacement of the UV filter for emitted fluorescence light (690–730 nm, minimum bandwidth 40 nm). The camera is operated at a maximum frame rate of 50 fps, resolution 1936×1216 pixels. The exposure is set at 28 ms with a gain setting of 3000. Data outputs include a series of 101 images captured over 2 s, including the 1 s saturating flash. The images are saved as bin files along with a metadata file that includes the field location and timestamp of the measurement.

3 Methods

3.1 High-Throughput Chlorophyll Fluorescence Imaging of Experimental Field Trials for Dark-Adapted Measurements of F_v/F_m

Procedures for chlorophyll fluorescence imaging in controlled environments are described comprehensively in [10]. The focus here is on the unique aspects of chlorophyll fluorescence imaging in field environments at a scale that accommodates experimental plantings of diversity and mapping panels (~300–900 field plots). While protocols and methods will vary among imaging systems and for different user objectives, several common factors need to be addressed when taking measurements of F_v/F_m on dark-adapted plants in field trials.

1. Plants must be in a fully dark-adapted state prior to the measurement. Restricting scan hours for measurements on dark-adapted plants to nighttime hours 2 h after sunset and before sunrise ensures that plants will be in a dark-adapted state. The intensity of the saturating flash is such that diffuse light disperses to neighboring plants (Fig. 2). Subsequent measurements on plants in the vicinity of a light flash will need to be far enough away to ensure that the measured plants remained fully dark adapted (*see Note 1*).
2. Trigger a saturating flash of light for a duration of 0.8–1 s at an intensity sufficient to saturate the plant photosystems of field-grown plants across the entire imaged area. The illuminating light source needs to provide an irradiance of $>4000 \mu\text{mol}/\text{m}^2/\text{s}$ to capture an accurate measurement of F_m [9, 10]. Results from tests adjusting the illumination intensity in our field scanner system indicate that irradiance of $3500 \mu\text{mol}/\text{m}^2/\text{s}$ is not sufficient, as expected. However, $4200 \mu\text{mol}/\text{m}^2/\text{s}$ is also not quite adequate (*see Note 2*).
3. The saturating flash needs to be synchronized with the camera images and with the programmed robotic motion between measurements. This engineering challenge in the field scanner system is addressed by adjusting the timing of the first image frame to the LED trigger function and programming motion functions that allow for a few seconds delay after the measurement and before movement to a new plot. The requirement for a rapid high-intensity saturating flash creates associated technical challenges in capturing the minimal level of fluorescence (F_0) at exactly the time just after the excitation light is applied [1, 7]. Moreover and perhaps unique to a field setting is that there are sources of background nighttime light that cannot be controlled, for example, moonlight or diffuse light from parking lot lamps (*see Note 3*).
4. A field imaging system needs to be robust against a wide range of environmental conditions. Supplementing the primary components with climate control notably adds to the size and

weight of the complete system. An air conditioner (KTS Hengstberger GmbH) that is able to cool as well as heat is incorporated within the interior chamber above the LED panel that houses the camera and electronics to provide climate control within ambient temperatures in the range from 0 to 50 °C. *See Note 4.*

3.2 Evaluation of the Influence of Variable Plant Heights

Given that the illuminating flash and the fluorescence camera remain at a constant height throughout a data collection scan, it is important to determine the influence of variable plant heights on the data outputs. Differences in plant heights can potentially influence F_v/F_m values as the intensity of the saturating pulse is decreased with increased distance from the light source to the plant target. Additionally, the ability to detect emitted photons may decrease with increased distance from the plant target. To evaluate the influence of variable distances from the plant target to the imaging system, a comparison test can be conducted taking repeat measurements over time on target plots of known height while adjusting the height of the fluorescence camera system. To maintain plants in a dark-adapted state, the measurements need to be separated by spatial distance (>4 m) or by time (~30 min or more). Test results completed with an illumination intensity of 4900 $\mu\text{mol}/\text{m}^2/\text{s}$ suggest that measurements of mean F_v/F_m on durum wheat at heading stage show a slight decreasing trend as the distance from the plant target to the imaging system increases (simulating a decrease in plant height) in the range from 70 to 140 cm. However, the differences in F_v/F_m are not significant ($p > 0.05$) between standard operational distance (80 cm) compared to distances ranging from 70 to 140 cm (Fig. 3). It is likely that the influence of variable plant heights differs across species with different leaf angles and plant architectures.

4 Notes

1. Preliminary tests on our system indicate that 4 m separation between measurements is sufficient.
2. We find that 4900–7000 $\mu\text{mol}/\text{m}^2/\text{s}$ appears to be sufficient at the standard operating distance from the target.
3. In order to screen out background light, the camera system is synchronized to trigger a single frame before the LED light to establish a baseline intensity in the absence of the red flash. The baseline correction in our system is completed within the camera before images are saved.
4. The instruments need to be protected not only from the external environment but also from internal heating in the electronics and LEDs. Additionally, protection from dust and air

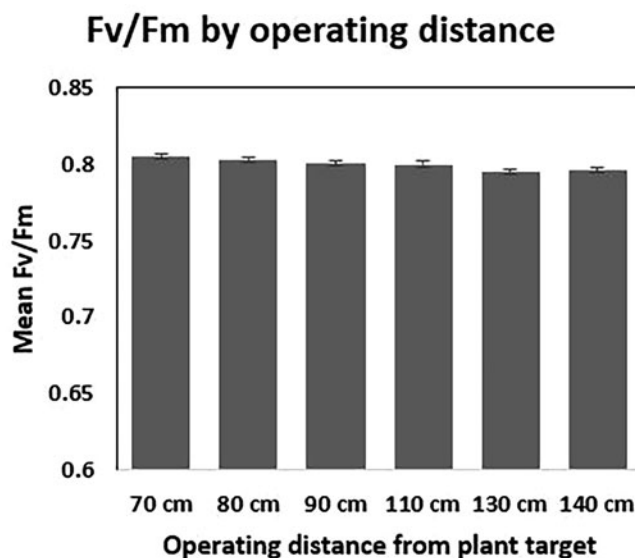


Fig. 3 Graph of mean F_v/F_m as a measure of the quantum efficiency of the PSII system for durum wheat variety Tiburon at heading stage, comparing control plots at a standard operating distance of 80 cm to plots measured at variable operating distances between 70 and 140 cm to simulate variable plant heights. Error bars represent $\pm$ SD

particulates is needed. The air conditioning unit has separated ventilation circuits to allow the clean air inside the chamber to recirculate without mixing with the outside air.

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